

Flex-Face: Privacy-Protect Face Recognition via Flexible Identity-Consistent Face Anonymization

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Abstract—Along with the rapid advancement of automated face recognition technology for societal security services, however, it has also raised privacy concerns about the exposure of private individuals' facial data under the ubiquitous camera collection. Thus, a notable paradox arises: societal and technological development needs a robust face recognition ecosystem for public safety and convenience, yet most people are reluctant to publicly release their facial data, causing a so-called Privacy-Utility trade-off. Numerous existing methods attempt to balance privacy and utility, yet often struggle with the dilemma of prioritizing one at the cost of the other, either suffering from low visual quality or unsatisfactory recognition performance while generating uncontrollable privacy-protected images. To tackle these challenges, this paper proposes a novel, flexible, and controllable paradigm for accurately recognizing privacy-preserving, high-visual-quality faces, aiming to achieve a balance between privacy preservation and visual utility. For accurate identity recognition, we introduce an Identifiable Face Anonymization module to hierarchically incorporate identity information into the privacy-protected images, cooperating with an identity recognition network. For privacy preservation, we design an identity-consistent face anonymization framework supporting the selection of diverse human-preferred identities for privacy preservation. This framework facilitates accurate face identification while preserving non-identity facial attributes, such as expression and pose, and achieving flexibility and controllability

in generating diverse human-preferred anonymized images. Comprehensive experimental results across various datasets validate the effectiveness of the proposed model from both quantitative and qualitative perspectives.

Index Terms—Face recognition, privacy protection, image watermarking, face anonymization.

I. INTRODUCTION

FACE recognition has gained great attention and made significant advancements in recent years, and been applied in various applications such as authentication, access control, and criminal pursuit. However, extensive development of such technology has also raised growing concerns about privacy due to the excessive collection and potential abuse of individuals' facial data. Persistent debates concerning the privacy concerns and legal implications of facial recognition systems [1], [2] highlight a stark contradiction between societal utility demands and individual privacy rights. Considering the necessity for security enhancement and social convenience, automatic and precise face recognition services are indispensable. Conversely, individuals under camera surveillance universally aspire to protect their facial privacy from both intentional and unintentional human observation [3]. Therefore, it is desirable for facial recognition technology to safeguard facial privacy from human scrutiny, while simultaneously retaining its utility for effective identity recognition by machines. However, striking a balance between utility and privacy in facial recognition systems poses an inherent complexity, stemming from the enduring tension that exists between data analytics and information preservation.

In response to the importance of privacy protection during facial capture and recognition, some research is underway to address privacy concerns arising from face images. The proposed strategies encompass a diverse range of techniques, including image sensors integrated with privacy enhancement [4], [5], adversarial methodologies for obfuscating recognition systems [6], [7], [8], generative face anonymization methods [8], [9], [10], face templates protection [11], [12], and secure photo sharing [13]. Most prevalent methodologies concentrate on eliminating visual information of individual identity through de-identification or anonymization, thereby rendering the protected images unrecognizable. Two branches of methodologies endeavor to balance facial privacy and recognition, as illustrated in Fig. 1. The first branch is called privacy-preserving face recognition (PPFR), which performs recognition in transformed domains such as encrypted codes or frequency components,

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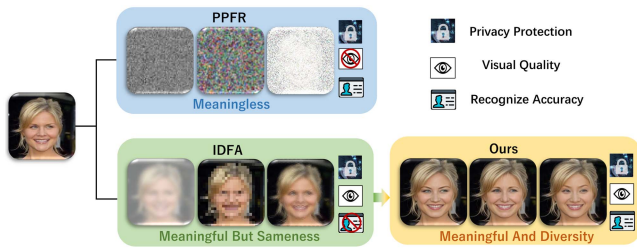


Fig. 1. Comparison of the methodologies for facial privacy and recognition. Privacy-Preserving Face Recognition methods focus on achieving superior recognition performance at the expense of sacrificing the visual quality of the protected images. Identifiable Face Anonymization methods obscure or anonymize faces and maintain visual interpretability, but suffer from reduced recognition accuracy. Consequently, the proposed approach enhances previous IDFA methods by achieving a good balance of privacy protection, image quality, and recognition accuracy.

ensuring high accuracy but suffering from low visual clarity. The other branch is called Identifiable Face Anonymization (IDFA), which alters the visual appearance naturally and keeps some identifiable features for recognition, while suffering from reduced recognition accuracy due to facial anonymity.

In recent studies, some efforts have been made to address the challenge of balancing privacy and utility. For example, PRO-Face [14] pioneered the concept of privacy-preserving recognizable obfuscation. It processes face images with preferred obfuscations, such as image blur, pixelation, and face morphing, and then utilizes a Siamese network to produce a protected image that appears visually similar to the obfuscated one (ensuring privacy) while remaining recognizable by machines (preserving utility). In addition, a subsequent approach PRO-Face C [15] integrates a client that transmits obfuscated images to the server and then performs identity recognition using a pre-trained model in conjunction with privacy-free complementary features. Recently, AIDPro [16] proposes a face privacy protection model by incorporating conditional messages to guide identity transformation for privacy protection, achieving utility-preserved privacy protection and robustness-enhanced image authentication. However, these methods have two main limitations: 1) Utility-privacy balancing: Considering that there exists a utility-privacy trade-off, it is hard to maintain visual clarity and high recognition accuracy simultaneously. 2) Limited flexibility: Most previous approaches always generate an objective privacy image for each input sample, lacking the flexibility to integrate human preference.

To solve the above-mentioned problems, we propose a flexible privacy-protective face recognition method via identity-consistent face anonymization to solve the privacy-utility paradox. Our model is derived from the following two aspects: On the one hand, considering the tradeoff between the visual appearance and recognition accuracy, motivated by the watermarking technique, we introduce a novel identity injection method to incorporate identity information hierarchically into the anonymized face images for natural visual generation and high recognition accuracy. On the other hand, in order to enhance the flexibility, we design an identity-consistent face anonymization framework that can incorporate optional diverse

human-preferred identities for privacy preservation, while still being accurately recognized by our recognizer. The contributions of this paper are summarized as follows:

- 1) We introduce Flex-Face, a new paradigm for identity-consistent face anonymization, which reframes identity-preserving recognition as an identity embedding and decoding problem, enabling strong visual anonymization with reliable recognition.
- 2) We propose a hierarchical identity injection mechanism that embeds identity information for robust recognition in a distributed manner, allowing recognition to remain stable even when encountering large facial appearance modifications.
- 3) We develop an identity-consistent anonymization framework that supports diverse, user-specified anonymization identities by decoupling visual appearance from recognition, achieving flexible and controllable anonymization.
- 4) Comprehensive quantitative and qualitative experiments across multiple image datasets showcase the effectiveness and flexibility of the proposed approach to protect privacy and maintain utility.

II. RELATED WORK

A. Face Anonymization

Face anonymization has become a popular research task with the aim of morphing facial appearance to protect face privacy, and it has been studied for a long time. In the early stages of face anonymization development, traditional methods based on basic image processing techniques, such as image blurring, masking, or other image transformations, were used to remove or obfuscate face images to protect privacy information directly. However, these traditional image processing-based face anonymization techniques suffer from the poor visual quality of the generated faces, containing blurring and artifacts. With the development of deep learning models, more and more researchers are working on deep learning-based face anonymization methods. Among those deep learning methods, Generative Adversarial Networks (GANs) [17] have been extensively studied, and a large number of face anonymization methods leverage GANs to edit the facial appearance to achieve anonymization. For example, Sun et al. [18] propose an inpainting method to generate anonymized faces, ensuring the transformation of the original face. Moreover, DeepPrivacy [19] utilizes Conditional GAN (CGAN) [20] with background and pose annotation as inputs to guide the generation of realistic anonymized faces. Recently, Cao et al. [21] proposed a personalized and invertible de-identification framework where a user-specific password and an adjustable parameter are used to control the direction and degree of identity variation. Furthermore, face swap techniques can also be applied for anonymization. For example, Sun et al. [22] proposed a face swap-based method for obfuscation, which combined a data-driven model with a parametric face model. Despite recent progress in face anonymization, most existing methods primarily focus on preserving recognition utility for downstream tasks such as face detection, analysis, and tracking.

Therefore, performing secure face recognition on anonymized images remains a challenging and underexplored problem.

B. Identity-Preserving Facial Anonymization

Identity-preserving Facial Anonymization methods aim to preserve specific identifiable characteristics within anonymized visual content. Specifically, IDFA endeavors to anonymize the facial appearance (to the human eye) while preserving the identity recognition capability (by machines). Recently, newly emerging studies have focused on this specific problem. Li et al. [23] first proposed identity-preserved face camouflage, which disentangles facial features into appearance and identification codes and replaces the former with another target. The same authors later proposed two more advanced approaches [24], where an identity-aware region discovery module is devised to determine the facial attributes sensitive to humans but less crucial to machines, based on which, facial anonymization is performed via the conditional GAN. Yuan et al. [14] proposed PRO-Face, a generic framework for recognizable face obfuscation inspired by image steganography. Accordingly, Wang et al. [25] devised a novel scheme via virtual identity transformation based on disentangled facial representation, which guarantees strong privacy and high identifiability. Zhang et al. [26] provide tailor-made face privacy protection for individuals via class-wise targeted universal adversarial perturbations. Pu et al. [27] propose a Privacy Minimization Transformation method that can process the private facial data based on the shallow network of the face recognition model to obtain the obfuscated data. Li et al. [28] employ semantic attribution maps to precisely obfuscate sensitive attributes (e.g., age, gender) while retaining identity features for matching. However, as anonymization inherently opposes recognition, the recognition performance of IDFA approaches typically struggles to achieve satisfactory recognition accuracy while ensuring good privacy protection. Building upon the foundation laid by pioneering researchers, our approach strives to attain a harmonious balance of robust visual privacy and optimal recognition utility.

C. Image Watermarking

Image watermarking is a technique that involves embedding specific information within a carrier image and subsequently retrieving this hidden content from the resulting stego image. Initially, these technologies relied on traditional image processing methods, including least significant bit (LSB) modifications [29], [30], [31] and Fourier transforms [32]. Recently, advancements in deep learning have led to the proposal of numerous watermarking methods, which exhibit superior watermarking capacity and robustness against image perturbations. For instance, Zhu et al. [33] introduced an end-to-end auto-encoder architecture, comprising an encoder, a noise layer, and a decoder, with an additional discriminator to ensure realistic visual quality for watermarking purposes. Ahmadi et al. [34] proposed a framework capable of operating in various domains, such as the Discrete Cosine Transform (DCT) domain, utilizing a residual structure for the encoder and incorporating a strength factor to control the intensity of watermark patterns within the

image. While the end-to-end framework facilitated the joint training of the encoder and decoder, the differential constraints of the noise layer hindered its practical application. Consequently, a two-stage separable deep learning framework [35] was proposed for practical watermarking. In the first stage, the encoder and decoder are initialized without the noise layer. In the second stage, the decoder is independently enhanced to accommodate non-differential distortions. Despite advancements in watermarking, the robustness of watermarking models remains a challenging problem. Supposing the identity information is directly embedded into the watermark of an anonymized face for face recognition, it remains an essential problem to obtain lower bit error rates under various kinds of perturbations in secure recognition. To overcome this issue, we develop a hierarchical identity injection method to incorporate identity information into facial images.

III. METHODS

A. Overview and Problem Formulation

We propose a novel paradigm for identifiable face anonymization methods, which aims to enable accurate machine recognition on visually interpretable but identity-altered facial images, achieving a balanced trade-off between visual privacy and recognition utility. Unlike PPFR methods, the proposed IDFA framework explicitly targets human-interpretable anonymized faces, where identity-specific cues are altered while non-identity attributes such as facial expression, pose, and structural characteristics are preserved. An overview of our model is shown in Fig. 2, highlighting a framework that performs identity recognition in an intuitive and straightforward manner: 1) Identity-aware facial embedding hierarchically injects the encrypted identity information into the original face image; 2) Identity-consistent face anonymization module anonymizes the input embedded faces to protect visual privacy for human eyes. 3) The identity recognizer recognizes the anonymized images and thus achieves high recognition accuracy by machine.

Two critical aspects of our model, namely privacy protection and identity recognition, can be formulated as follows:

a) *Privacy protection*: Given an input face image, denoted as $\mathbf{X}$, we obfuscate it using a face anonymization operator F , which allows for a manually specifiable identity shift, formulated as:

$$\mathbf{X}^A = F(\mathbf{X}, \mathbf{I}_{target}), \quad \text{s.t. } \mathbf{X}^A \in \mathcal{V}, \quad (1)$$

where $\mathbf{X}^A \in \mathcal{V}$ denotes the anonymous image constrained within the natural face manifold $\mathcal{V}$, ensuring its human-interpretable. Unlike PPFR methods that may rely on abstract noise or latent perturbations, our framework explicitly employs a target visible reference face $\mathbf{I}_{target}$, which can be either sampled from a reference set or generated by a synthesis model. This formulation ensures that the anonymization process results in a visually realistic face that maintains semantic utility for human observers, rather than an uninterpretable digital representation.

b) *Identity Recognition*: The Identity Recognizer is denoted as R , which takes the anonymous image $\mathbf{X}^A$ or the original face

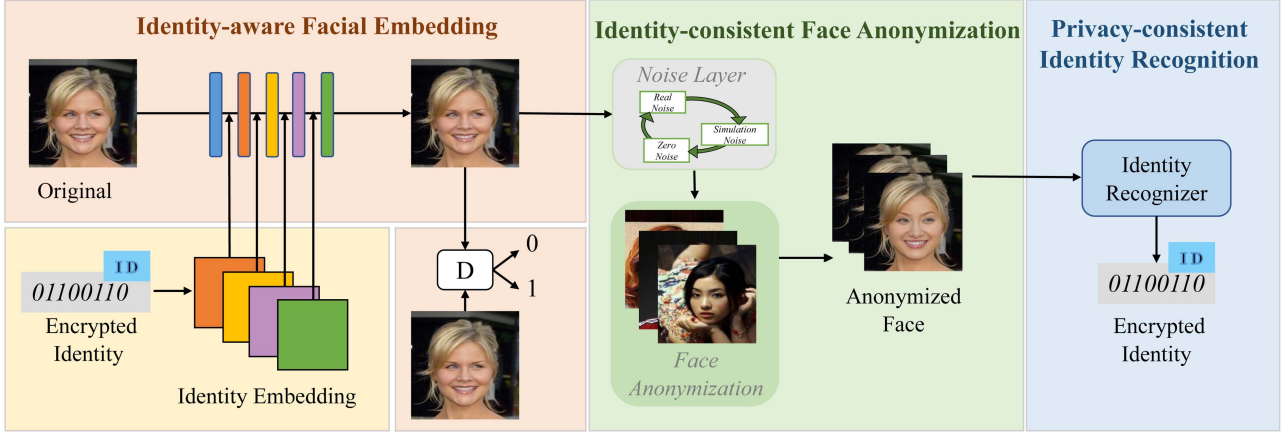


Fig. 2. Overall framework of the proposed model: The original input image is first injected with an encrypted identity embedding, and then an identity-aware facial image is generated constrained with an adversary discriminator. Subsequently, we employ a noise layer to enhance robustness and leverage a face anonymization module to generate anonymous faces. To this end, we input the anonymous faces into an identity recognizer for recognition.

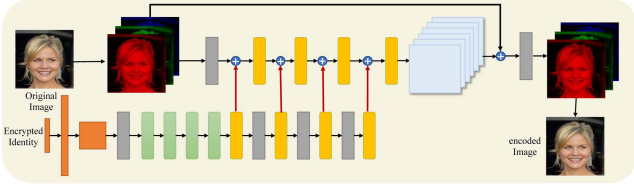


Fig. 3. Identity injection model: In order to achieve high recognition accuracy, the encrypted identity embedding is hierarchically incorporated into the image in the latent space and then mapped into the observation space.

image $\mathbf{X}$ as input and outputs the encrypted identity c , stated as:

$$c = R(\mathbf{X}) = R(\mathbf{X}^A), \quad (2)$$

where c is the encrypted identity. Noting that the target images $\mathbf{I} \in \Omega$ can be diverse, but all the anonymized images $\mathbf{X}^A$ keep the same recognizable identity as the input image $\mathbf{X}$.

B. Identity-Aware Facial Embedding

The most critical and challenging part of identity-preserving facial anonymization is how to achieve accurate identity recognition by a machine after face anonymization. Considering that the face anonymization step always makes modifications and removes identity-related characteristics, the objectives of privacy preservation and recognition utility are somewhat conflicting. Thus, in this paper, we introduce a new way of embedding identity-related information into the whole image, including both the facial regions and the background regions. Therefore, even though the face information changed, the model remains capable of recognizing the original identity of the anonymized facial image. Our identity-aware facial embedding model involves three parts, specified as follows:

a) *Encrypted Identity Embedding*: To enhance the encoding of the identity information, we incorporate an encrypted identity code processor to manage the identity message and subsequently provide the identity feature maps for identity injection. First, we generate an encrypted identity code $C = [C_1, \dots, C_k, \dots, C_K]$

for each identity through random sampling, where $C_k \in \{0, 1\}$. To facilitate the embedding process into the image, we reshape the encrypted identity code from $\mathbf{R}^K$ into $\mathbf{R}^{m \times n}$, where $K = m \times n$. Then the reshaped encrypted identity code is embedded via 4 up-sampling convolution layers Ψ , which finally yields the same shape as the image, and we denote the l^{th} identity embedding as $\Psi(C)^l$. Subsequently, the hierarchical identity embeddings are then incorporated into the images via an identity injection module.

b) *Identity Injection*: In this module, we aim to incorporate the hierarchical identity embeddings into the original image in the hierarchical latent space. As shown in Fig. 3, the l^{th} identity embedding $\Psi(C)^l$ is hierarchically concatenated with the encoded image features at four layers respectively. Then we concatenate the initial features with a residual connection and the embedded features through a 1×1 convolutional layer. Denote this hierarchical embedding process as Π , we generate an identity-aware embedded image $\Pi(\mathbf{X}, \Psi(C))$ where each pixel encompasses identity information.

c) *Adversary Discriminator*: Furthermore, we introduce an adversary discriminator to guarantee the consistency and visual quality of the output image after injecting identity embeddings. Specifically, we utilize a discriminator D composed of 4 convolution layers and a classifier to distinguish the embedded image and the original image, while encouraging the identity injection network to make them indistinguishable. The objective can be stated as

$$\mathcal{L}_{em} = \min_D \max_{\Psi, \Pi} \log(1 - D(\Pi(\mathbf{X}, \Psi(C)))) + \log(D(\mathbf{X})). \quad (3)$$

C. Identity-Consistent Face Anonymization

a) *Noise Layer*: To enhance the robustness against various distortions, we incorporate a noise layer. For each mini-batch during training, one noise mechanism is randomly chosen from a real JPEG compression noise $\mathcal{N}_r$, a differential simulation noise $\mathcal{N}_s$, and a noise-free layer $\mathcal{N}_z$. The functionality of each mechanism can be demonstrated as:

Real JPEG Noise Layer $\mathcal{N}_r$: The real JPEG noise is used for the subsequent network to learn robust features to make correct identity recognition even with real JPEG compression. Note that JPEG compression needs to quantize the discrete cosine transform (DCT) coefficients of image blocks, which is non-differential, thus the gradients propagated back [35], [36] to the encoder become zero.

Simulation Noise Layer $\mathcal{N}_s$: The simulation noise is generated as an alternation of the real JPEG compression to improve the robustness of the model further. Note that the simulation noise of JPEG compression provides the gradient that can be propagated back to the encoder so that the whole network can be jointly updated.

Zero Noise Layer $\mathcal{N}_z$: We set a probability with no noise to ensure the decoding ability without JPEG compression.

For each training batch, one of the above three mechanisms is randomly sampled and conducted on the identity-aware encoded facial image $\mathbf{X}^e$, which is then passed to the face-swapping module, stated as:

$$\mathbf{X}^n = \begin{cases} \mathcal{N}_r(\mathbf{X}^e), & f(\eta) = 2 \\ \mathcal{N}_s(\mathbf{X}^e), & f(\eta) = 1 \\ \mathcal{N}_z(\mathbf{X}^e), & f(\eta) = 0 \end{cases} \quad (4)$$

where η is uniformly sampled from the continuous interval $[0,3)$, and $f(\cdot)$ refers to the floor function mapping η to a discrete index in $\{0,1,2\}$, which results in equal selection probability for each noise layer. By incorporating the noise layer, the facial embedding module is motivated to generate robust embedding, and the identity recognizer is motivated to recover the encrypted identity even with noisy input, preventing the ‘overfitting’ problem.

b) Face Anonymization: In contrast to previous identity-preserving facial anonymization methods that either cannot generalize to diverse anonymization identities or fail to preserve attributes like facial expression and gaze direction of the original images, our model aims to transfer arbitrary identities while preserving the attributes of the original images. Motivated by [37], we design a face anonymization framework that is capable of generating anonymous images with diverse identities of target images $\mathbf{I} \in \Omega$ while keeping the same recognizable identity as the input image $\mathbf{X}$. Our approach provides high flexibility, allowing for the incorporation of diverse, user-preferred identities to enhance privacy preservation. As shown in Fig. 4, our face anonymization process is composed of three parts and is constrained by two objectives, specified as follows:

Anonymization Encoder: We first employ an encoder AE consisting of 4 down-sampling convolution layers to embed the noised image $\mathbf{X}^n$ into a latent space for deriving facial anonymization. In this step, both the identity and attribute of the original face are extracted. In the latter module, the modification should be focused on the anonymization of the original facial part while keeping the original attributes (e.g., expression, posture, lighting, etc.) unchanged.

Identity Transformation: This step incorporates the identity information of the target image $\mathbf{I}$, which could be either manually assigned or randomly sampled from Ω , into the original face for anonymization, while keeping the original attribute information unchanged. Thus, we employ an identity extractor Φ_{id} on the

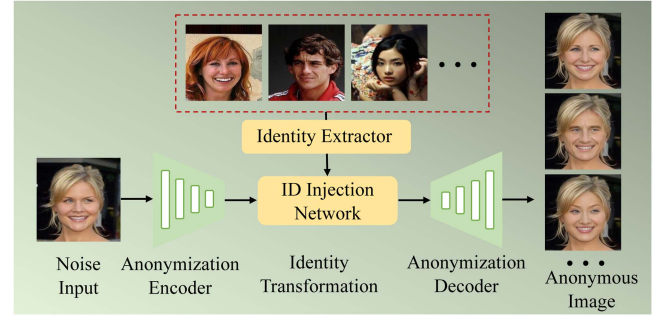


Fig. 4. The network of face anonymization. The noise input is first encoded with an anonymization encoder, then anatomized in the latent space and transformed into another identity, and finally decoded with an anonymization decoder to generate anonymous images.

target image $\mathbf{I}$ to extract its identity vector, then utilize an ID injection network Θ consisting of residual CNNs to inject the identity information into the original face in the latent space, stated as $\Theta(AE(\mathbf{X}^n), \Phi_{id}(\mathbf{I}))$. Note that during this step, the attribute of the original face should remain unchanged, and the anonymous face should be only confused to human eyes but still recognized by the recognizer.

Anonymization Decoder: Then we input the transferred features into the anonymization decoder AD consisting of 4 up-sampling convolution layers to obtain the anonymous face image by $\mathbf{X}^a = AD(\Theta(AE(\mathbf{X}^n), \Phi_{id}(\mathbf{I})))$. Considering the training incorporates target images of various identities, the decoder is not tied to any particular identity. Thus, our anonymization decoder concentrates on reconstructing the image given the transferred features, not limited to the specific facial recovery. This allows our face anonymization model to be flexible and adaptable for any identity.

During the whole training pipeline, two constraints on identity and attributes are present: First, to ensure the identity knowledge of the target image $\mathbf{I}$ has been successfully transferred into the input noised facial image $\mathbf{X}^n$, an identity matching loss is employed as $\mathcal{L}_{id} = 1 - \frac{\Phi_{id}(\mathbf{I}) \Phi_{id}(\mathbf{X}^a)}{\|\Phi_{id}(\mathbf{I})\|_2 \|\Phi_{id}(\mathbf{X}^a)\|_2}$. Second, to encourage the anonymous face to keep the attributes of the original face, we utilize a feature matching loss, motivated by [38], with a discriminator Dis to constrain the similarity of multiple layers of features of the input image and the anonymous image, stated as $\mathcal{L}_{att} = \sum_{l \in L} \|\text{Dis}(\mathbf{X}^n)^{(l)} - \text{Dis}(\mathbf{X}^a)^{(l)}\|_1$. To constrain the attribute consistency and consider that the attribute is high semantic information mainly embedded in deep features, the feature matching loss only constrains the matching between the last few layers of the Discriminator.

D. Privacy-Protective Identity Recognition

Our model endeavors to preserve privacy by anonymizing facial appearance while maintaining the original identity recognizable by the machine, specified as the following requirements: i) Privacy Preservation: As a fundamental requirement, it should be impossible for humans to retrieve the identical visual representation of the original face from the anonymized image. ii) Identity Recognition: Given the privacy-protected anonymous

image, the machine should be able to recognize identity via a recognition model accurately. However, it is challenging for a regular recognition model to deduce the original face identity from an anonymous image.

To facilitate recognition of the original identity, a specialized identity recognition module Φ_{rec} is introduced to investigate the intrinsic identity information through the whole image, which is embedded with the identity-aware facial embedding module. Motivated by [36], in the decoder, the anonymous image $\mathbf{X}^a$ is first input into a 3×3 Conv-BN-ReLU layer and then mapped with several Squeeze-and-Excitation (SE) blocks to learn features from the frequency domain. Subsequently, a 3×3 convolution layer is employed to convert the multi-channel tensor into 1 channel and then get the identity code. Note that during the training stage, we encourage the model to reconstruct the same identity code of the anonymous image as the original image, stated as $\mathcal{L}_{rec} = \sum_{k=1}^K (\Phi_{rec}(\mathbf{X}^a)_k - C_k)^2$, where k refers to the k^{th} element of the recognized/original identity code. It should be noted that frequency-domain information is modeled implicitly through convolutional feature responses and channel-wise attention, rather than via explicit frequency transformations such as DCT. In particular, convolutional features naturally contain mixed frequency components, and the SE blocks enable the decoder to emphasize frequency-sensitive identity cues while suppressing appearance-related variations. Extensive experiments demonstrate that our identity recognition module can correctly recognize identities in anonymized images even under strong obfuscation and significant modifications.

E. Overall Objective

To this end, we train the whole network with a total objective loss by optimizing the network parameters. The total training loss function is computed as the weighted sum of the aforementioned loss functions:

$$\mathcal{L} = \lambda_{em}\mathcal{L}_{em} + \lambda_{id}\mathcal{L}_{id} + \lambda_{att}\mathcal{L}_{att} + \lambda_{rec}\mathcal{L}_{rec}, \quad (5)$$

where λ_* refers to the corresponding weights of different losses. Note that $\mathcal{L}_{em}$ is utilized to maintain the original visual content, $\mathcal{L}_{id}$ ensures the target identity information is well transferred into the original image for privacy protection, $\mathcal{L}_{att}$ constrains the attribute consistency of anonymous image for visual utility, and $\mathcal{L}_{rec}$ encourages high recognition accuracy by machine. In this way, our model is encouraged to achieve a good balance between privacy protection, visual utility, and recognition accuracy, while acquiring flexibility to incorporate diverse human-preferred identities for privacy preservation.

IV. EXPERIMENTS

In this section, we conduct comprehensive experiments to evaluate the effectiveness of the proposed method in terms of both privacy protection and utility. Using full-reference and no-reference metrics, we quantify image quality and assess the visual differences between anonymized and original images to measure the impact of anonymization on privacy. Additionally, we analyze face recognition accuracy across different scenarios

to confirm that our method maintains high accuracy while ensuring privacy protection. We also evaluate the model's robustness under various perturbation conditions to verify the stability of anonymized images in practical applications.

A. Datasets

We utilized three publicly available facial datasets in our experiments. The Labeled Faces in the Wild (LFW) dataset contains over 13,000 images spanning 5749 identities, making it well-suited for face recognition and verification tasks. The CelebFaces Attributes Dataset (CelebA) includes more than 200,000 celebrity images annotated with 40 attribute labels, providing a versatile dataset for multi-task learning. Finally, the VGG-Face2 dataset, containing over 3,000 identities and nearly 900,000 images, is specifically designed for training deep learning models.

Our Flex-Face model is trained on the CelebA dataset and evaluated on all three datasets mentioned above. For face verification on the LFW dataset, we follow the standard protocol provided by LFW [39], which includes 6,000 image pairs: 3,000 matching pairs (same identity) and 3,000 non-matching pairs (different identities). For face verification on the CelebA and VGG-Face2 datasets, we mimic the LFW benchmark by randomly selecting 3,000 matching pairs and 3,000 non-matching pairs from each dataset. For face recognition tasks, we use 10% of the CelebA dataset as a test set and select 10,000 images from the VGG-Face2 dataset as an additional test set, evaluating performance using the Rank-1 accuracy.

Our baseline comparisons include Password-Conditioned [40], CIAGAN [9], DeepPrivacy [19], DeepPrivacy2 [41], and PRO-Face [14], all of which are deep learning-based facial anonymization techniques. Notably, CIAGAN, DeepPrivacy, and DeepPrivacy2 lack identity recognition capabilities, serving solely for anonymization. In contrast, Password-Conditioned, PRO-Face, and our proposed method preserve identity information during anonymization. For consistency, we conduct all comparisons on the same test set using each method's respective pre-trained model.

B. Implementation Details

The Adam optimizer is used for model training, with a learning rate of 0.001 and a batch size of 16. Data augmentation is applied during training to improve model generalization. For preprocessing, all images are resized to a resolution of 256×256 . The identity encoding module employs a 3×3 ConvBNReLU layer and SE block structure, commonly used in feature extraction tasks, as the basic block to better capture the original image's features. Multi-scale and multi-channel embeddings are sampled for more effective identity encoding. In the face anonymization module, the pre-trained SimSwap model is utilized for random face-swapping. SimSwap enables efficient identity transfer from any source face to any target face while preserving the target's attributes. For identity extraction, the anonymized image passes through a ConvBNReLU layer and four SE blocks, followed by a 3×3 convolutional layer that converts the multi-channel tensor to a single-channel output,

TABLE I

FULL-REFERENCE OBJECTIVE MEASUREMENT OF PRIVACY BY SIMILARITY BETWEEN PROTECTED AND ORIGINAL IMAGES (ON LFW)

LFW	LPIPS $\uparrow$	SSIM $\downarrow$	MAE $\uparrow$	RMSE $\uparrow$
Password-conditioned [40]	0.196	0.809	0.036	0.055
DeepPrivacy [19]	0.074	0.899	0.022	0.055
DeepPrivacy2 [41]	0.071	0.897	0.019	0.058
PRO-Face [14]	0.228	0.716	0.043	0.061
Ours	0.389	0.478	0.112	0.186

TABLE II

FULL-REFERENCE OBJECTIVE MEASUREMENT OF PRIVACY BY SIMILARITY BETWEEN PROTECTED AND ORIGINAL IMAGES (ON CELEBA)

CelebA	LPIPS $\uparrow$	SSIM $\downarrow$	MAE $\uparrow$	RMSE $\uparrow$
Password-conditioned [40]	0.201	0.799	0.038	0.058
CIAGAN [9]	0.343	0.532	0.100	0.169
DeepPrivacy [19]	0.099	0.867	0.029	0.069
DeepPrivacy2 [41]	0.091	0.866	0.026	0.071
PRO-Face [14]	0.228	0.716	0.043	0.061
PRO-Face C [15]	0.440	0.617	—	—
Ours	0.474	0.483	0.106	0.175

which is then reshaped to obtain identity encoding information. The loss weights are determined through empirical tuning. Specifically, we first normalize the magnitudes of different loss terms to ensure comparable gradient scales during early training. Then, the relative weights are adjusted according to their functional roles and task priorities, where identity preservation and message recovery are treated as primary objectives, while adversarial imperceptibility serves as a regularization term. The model is trained for 100 epochs, with the best-performing model on the validation set selected for testing. All experiments are conducted using the PyTorch framework on two NVIDIA RTX 2080 Ti GPUs.

C. Image Quality Evaluation

a) Full-Reference Metrics: Full-reference image quality evaluation requires comparing the original image with the anonymized image to assess quality based on the differences between them. To measure structural similarity, we employ the Structural Similarity Index Measure (SSIM) and Learned Perceptual Image Patch Similarity (LPIPS), with values in the range of [0, 1]. Lower SSIM and higher LPIPS scores indicate greater differences between the original and anonymized images, suggesting stronger privacy protection. Additionally, Mean Absolute Error (MAE) and Root Mean Square Error (RMSE) measure the average absolute and root mean square differences between the generated and original images; larger values in these metrics indicate greater privacy protection.

The experimental results on the LFW dataset, shown in Table I, indicate that our method outperforms other methods in terms of LPIPS, SSIM, MAE, and RMSE. Furthermore, Table II demonstrates that our method achieves the highest performance across all four metrics on the CelebA dataset. Our approach demonstrates superior privacy preservation, as evidenced by the LPIPS scores of 0.389 and 0.474 on the

TABLE III

NO-REFERENCE OBJECTIVE MEASUREMENT OF PRIVACY BY SIMILARITY BETWEEN PROTECTED AND ORIGINAL IMAGES (ON LFW)

LFW	DBCNN $\uparrow$	MUSIQ $\uparrow$	BRISQUE $\downarrow$	NIQE $\downarrow$
Password-conditioned [40]	0.397	29.871	17.786	6.332
CIAGAN [9]	0.429	38.168	25.644	6.023
DeepPrivacy [19]	0.293	39.337	37.392	6.414
DeepPrivacy2 [41]	0.357	44.839	30.834	5.641
PRO-Face [14]	0.271	30.543	32.830	5.241
Ours	0.623	60.521	17.056	5.112

LFW and CelebA datasets, respectively. These higher LPIPS values indicate a significant perceptual difference between the original and anonymized images, effectively enhancing privacy without compromising the quality of the anonymized faces. Additionally, the SSIM scores of 0.478 on the LFW and 0.483 on the CelebA further substantiate the structural alterations made to the facial features, ensuring robust privacy protection. At the same time, the higher MAE and RMSE values across both datasets reflect the notable pixel-level differences between the original and anonymized images, emphasizing the effectiveness of our method in safeguarding privacy. These results collectively show that our method effectively balances the trade-off between privacy protection and image quality, outperforming other approaches in both privacy-related metrics and visual integrity.

b) No-Reference Metrics: No-reference image quality evaluation, which does not require the original image, provides greater flexibility in assessments. Typically, this evaluation leverages the statistical characteristics of the image or deep learning models. DBCNN, MUSIQ, BRISQUE, and NIQE are all no-reference image quality metrics, each utilizing different methods to assess visual quality. DBCNN employs a dual bilateral convolutional neural network model, which assesses image quality by extracting image features and combining them with learned mapping relationships, making it well-suited for complex image distortions. MUSIQ integrates multiple task features within a multi-task learning framework to evaluate image quality from various perspectives, enhancing both the comprehensiveness and accuracy of the assessment. BRISQUE assesses images based on the statistical properties of natural scenes, emphasizing spatial domain information to effectively detect various types of distortions. NIQE evaluates distortion levels by analyzing image naturalness to assess quality. For DBCNN and MUSIQ, higher scores indicate better image quality, whereas for BRISQUE and NIQE, lower scores signify better quality.

The evaluation results on the LFW and CelebA datasets, as shown in Tables III and IV, further highlight the effectiveness of our method in both privacy protection and image quality. Specifically, our method achieves the best scores across the DBCNN, MUSIQ, BRISQUE, and NIQE metrics on the LFW dataset, significantly outperforming the other anonymization methods. This superior performance indicates that our approach not only provides strong privacy protection but also generates anonymized images that maintain high perceptual quality. Similarly, on the CelebA dataset, our method again leads across all

TABLE IV
NO-REFERENCE OBJECTIVE MEASUREMENT OF PRIVACY BY SIMILARITY
BETWEEN PROTECTED AND ORIGINAL IMAGES (ON CELEBA)

CelebA	DBCNN↑	MUSIQ↑	BRISQUE↓	NIQE↓
Password-conditioned [40]	0.448	31.797	14.128	6.365
CIAGAN [9]	0.536	38.027	21.503	6.807
DeepPrivacy [19]	0.332	41.690	34.087	6.618
DeepPrivacy2 [41]	0.446	49.512	25.621	5.491
PRO-Face [14]	0.371	34.516	30.134	5.360
Ours	0.652	63.284	11.927	5.334

TABLE V
FULL-REFERENCE OBJECTIVE MEASUREMENT OF PRIVACY BY SIMILARITY
BETWEEN PROTECTED AND ORIGINAL IMAGES (ON VGG-FACE2)

VGG-Face2	LPIPS↑	SSIM↓	MAE↑	RMSE↑
DeepPrivacy [19]	0.077	0.842	0.031	0.068
DeepPrivacy2 [41]	0.087	0.825	0.033	0.073
PRO-Face [14]	0.234	0.739	0.042	0.059
Ours	0.392	0.448	0.114	0.186

TABLE VI
NO-REFERENCE OBJECTIVE MEASUREMENT OF PRIVACY BY SIMILARITY
BETWEEN PROTECTED AND ORIGINAL IMAGES (ON VGG-FACE2)

VGG-Face2	DBCNN↑	MUSIQ↑	BRISQUE↓	NIQE↓
DeepPrivacy [19]	0.351	41.682	32.872	6.073
DeepPrivacy2 [41]	0.435	47.712	26.184	5.195
PRO-Face [14]	0.321	29.832	35.081	5.744
Ours	0.634	60.956	16.696	5.018

the evaluated metrics, reinforcing the consistency and robustness of our approach in preserving visual integrity while ensuring effective privacy. These results collectively demonstrate that our method not only ensures robust privacy protection but also generates high-quality anonymized images, which is crucial for real-world applications requiring both privacy preservation and visual quality.

Furthermore, to evaluate the generalization capability of our Flex-Face, we compare our method with DeepPrivacy, DeepPrivacy2, and PRO-Face on the VGG-Face2 dataset, computing both full-reference and no-reference metrics. In the full-reference metrics presented in Table V, Flex-Face achieves the highest LPIPS (0.392) and the lowest SSIM (0.448), together with higher MAE (0.114) and RMSE (0.186), suggesting stronger pixel- and structure-level perturbations. Table VI shows similar gains in no-reference metrics, where Flex-Face substantially improves DBCNN and MUSIQ while reducing BRISQUE and NIQE. These results confirm that Flex-Face generalizes well across datasets, achieving both stronger privacy protection and higher visual fidelity.

In summary, the significant improvements observed across various metrics demonstrate the superior performance of our Flex-Face in terms of privacy protection, image quality, and anonymization diversity. These results further verify the model's strong generalization ability and its capability to generate high-quality anonymized images while preserving privacy.

TABLE VII
FACE VERIFICATION PERFORMANCE. TAR@FAR(0.01) IS REPORTED
TO BE CONSISTENT WITH COMPETING METHODS

Methodology	LFW	CelebA	VGGFace2
Li et al. [24]	/	78.39	61.28
Li et al. [28]	/	82.00	89.00
IVFG [42]	/	92.90	93.30
PRO-Face [14]	97.52	93.71	78.38
PRO-Face C [15]	99.83	94.93	92.03
CanFG [43]	/	98.80	95.10
Wang et al. [44]	/	95.51	93.50
KFAAR [45]	98.40	98.90	/
Ours	100.00	100.00	99.23
Ours(AUC)	100.00	100.00	99.58
Ours(Rank-1)	100.00	100.00	99.93

D. Recognition Accuracy

To evaluate the recognition performance of our Flex-Face, we adopt the following metrics:

a) Face Verification Accuracy: For a given pair of face images, we determine whether they belong to the same identity. The model's performance on the verification task is evaluated by calculating the ratio of correctly classified pairs.

b) Rank-K Recognition Rate: In the test set, if the model correctly identifies the target identity within the top K predictions, it is considered a success. We compute the Rank-1 recognition rate for each identity.

For face verification, we use $\text{TAR@FAR} = 0.01$ as the evaluation metric to compare our method with existing identity-preserving anonymization methods. Although these methods adopt different strategies to alter the appearance of the original images, the results show that our approach achieves a significant advantage. The recognition performance is notably improved, achieving exceptionally high face verification accuracy across all three datasets. As shown in Table VII, our face verification accuracy reaches an impressive 100% on the LFW and CelebA datasets. Moreover, on the VGG-Face2 dataset, the accuracy is as high as 99.23%, representing a 7% improvement over PRO-Face. This improvement is attributed to the identity embedding and extraction scheme we developed, highlighting the superiority of our approach.

We further validate the face recognition performance of our method across the three datasets by calculating the Rank-1 recognition rate using a Gallery and Query approach. Specifically, we first construct a Gallery for each individual, storing their identity labels and corresponding identity embeddings. We then generate a Query set using the anonymized images produced by the model. For each anonymized image in the Query set, we extract its identity embedding and compare it to the identity embeddings of all images in the Gallery. We compute the cosine similarity between the embeddings, rank the Gallery images based on similarity scores, and verify whether the identity label of the top-ranked image matches the true identity of the Query image. Additionally, on both the LFW and CelebA datasets, our Rank-1 recognition rate reaches 100%. On the VGG-Face2 dataset, our Rank-1 recognition rate achieves



Fig. 5. Visual results of different image distortions on anonymized images.

99.93%, demonstrating our method's strong generalization ability across different datasets.

E. Robustness of Anonymized Images

Considering that anonymized images may be degraded by compression, resolution reduction, and various perturbations during transmission or malicious processing, we further evaluate the robustness of our method under both distortion and reconstruction attack scenarios. Specifically, different levels of perturbations (e.g., noise, blur, compression, cropping, and low-resolution degradation) are applied to the anonymized images to simulate realistic transmission conditions. In addition, we conduct reconstruction attack experiments by employing image restoration models to recover potential identity-related information from anonymized images. These experiments aim to assess whether Flex-Face can maintain reliable identity recognition while preserving high visual quality and strong privacy protection under both common distortions and adversarial reconstruction attempts.

a) Robustness to Common Image Distortions: To evaluate the robustness of the proposed method under realistic image degradation conditions, we apply several common image distortions to the anonymized images, including Gaussian noise, Gaussian blur, JPEG compression, and pixel dropout. Specifically, we use the Albumentations library to assess the robustness of anonymized images, conducting tests on the LFW dataset. We select 6,000 images from 4,281 individuals as our test set. Fig. 5 shows the anonymized images alongside those subjected to various disturbances. The figure illustrates the impact of different types of image distortions on the anonymized images. It presents an anonymized image and its versions modified with Gaussian noise, Gaussian blur, JPEG compression, and pixel dropout. These visual examples allow us to evaluate the robustness of our model in maintaining recognizable identity information under various distortions.

To simulate the random noise that may occur in low-quality images and to enhance the model's robustness against such noise interference, we apply Gaussian noise with a mean of 0 and variances of 100, 200, and 300 to the anonymized images. Although the introduction of noise results in noticeable artifacts and a decline in visual quality, as shown in Table VIII, the impact of Gaussian noise on our model's face verification accuracy and Rank-1 recognition rate remains minimal. This can be attributed to the extremely low bit error rate in identity encoding extraction, which is nearly zero, leading to very high accuracy in face verification and recognition tasks.

For Gaussian blurring, we apply Gaussian kernels of varying sizes to the anonymized images to reduce details and obscure

TABLE VIII
ROBUSTNESS EVALUATION ON LFW

Perturbation	Parameter	BER	TAR(%)	Rank-1(%)	AUC
Gaussian noise	100	0	99.93	99.97	0.9999
	200	0.0012	99.87	99.90	0.9994
	300	0.0026	99.57	99.78	0.9989
Gaussian blur	21	0.0010	99.77	100.00	0.9997
	23	0.0085	99.03	100.00	0.9967
	25	0.0438	85.63	100.00	0.9488
JPEG	60	0.0018	99.77	99.91	0.9998
	40	0.0111	99.50	99.73	0.9983
	20	0.1026	83.57	91.40	0.9976
Pixel dropout	0.1	0.0010	99.83	99.91	0.9995
	0.3	0.0125	99.27	99.76	0.9987
	0.5	0.0446	96.93	99.52	0.9958

edges and textures. This approach aims to help the model cope with scenarios involving inaccurate focus and unclear images, thereby enhancing its generalization ability under blurred conditions. Since the degree of blurriness in Gaussian blurring is determined by both the kernel size and the standard deviation, we experiment with different kernel sizes (21, 23, 25) and standard deviations (1.0, 1.2, 1.4). Notably, when the Gaussian kernel size is 21 or 23, the face verification accuracy exceeds 99%, with a Rank-1 recognition rate of 100%. Although increasing the kernel size and standard deviation results in more pronounced blurriness—an unavoidable outcome—even with a kernel size of 25 and a standard deviation of 1.4, the face verification accuracy decreases to 85.63%. However, the Rank-1 recognition rate remains impressively high, demonstrating our Flex-Face's ability to handle unclear images effectively.

To simulate quality loss from high compression, we apply JPEG compression to the anonymized images at varying quality settings. Given that JPEG compression is a lossy method, high compression ratios lead to noticeable degradation in image quality and the introduction of compression artifacts, allowing us to evaluate the model's performance under low-quality conditions. We set the compression parameters to 60, 40, and 20, with lower parameter values corresponding to higher levels of compression. As the compression degree increases, it is inevitable that verification and recognition accuracies decline, and some errors appear in the extraction of identity encodings. Notably, when the anonymized images are compressed to 60% and 40%, both the face verification accuracy and the Rank-1 recognition rate exceed 99.5%. Even when the images are compressed to 20%, the Rank-1 recognition rate remains high at 91.4%, and the AUC consistently stays above 0.99. This demonstrates that our Flex-Face maintains excellent performance even under low-quality image conditions.

In the case of pixel dropout, we simulate image noise during transmission by randomly discarding certain pixels from the anonymized images and replacing them with a predefined value. This approach allows us to evaluate the model's robustness against interference with varying dropout probabilities. When the dropout rates are set to 0.1 and 0.3, both the face verification accuracy and the Rank-1 recognition rate remain above 99.20%. This robustness is attributed to the model's ability to embed a

TABLE IX
IMPACT OF IMAGE RESOLUTION ON ANONYMIZATION AND RECOGNITION PERFORMANCE

size	TAR(%)	AUC	Rank1(%)	BER
256*256	100.00	1.00	100.00	0.0011
128*128	100.00	1.00	100.00	0.0091
64*64	67.90	0.97	83.96	0.4994

30-bit identity encoding across the entire image, rather than limiting it to specific regions. Furthermore, even with a dropout rate of 0.5, where the error rate for identity encoding extraction reaches 10%, the face verification accuracy still achieves 96.93%, with the AUC consistently remaining above 0.99. This demonstrates the strong resilience of our Flex-Face against interference.

b) Robustness Across Different Image Resolutions: To simulate practical low-resolution scenarios encountered in real-world applications, we conduct experiments by resizing the anonymized images to different spatial resolutions before performing face recognition and identity decoding. Specifically, we conducted additional experiments by resizing input images to 128×128 and 64×64 , while keeping the network architecture unchanged. To simulate low-resolution surveillance scenarios, input images are downsampled and then upsampled back to 256×256 before being fed into the network. As listed in Table IX, the results show that the performance at 128×128 remains highly consistent with that at 256×256 , indicating that the proposed framework is robust to moderate resolution degradation. However, when the resolution is further reduced to 64×64 , we observe a performance drop in both recognition accuracy and identity decoding reliability.

This phenomenon is expected, as extremely low-resolution inputs inherently lack sufficient high-frequency facial details required for both face recognition and robust identity embedding extraction. Importantly, such degradation is not specific to our method but is commonly observed across face recognition and anonymization systems under severe resolution constraints. Nevertheless, the proposed approach maintains strong performance in practical surveillance-like resolutions (e.g., 128×128) commonly used in real-world systems.

c) Robustness Under Reconstruction Attacks: To evaluate the resilience of the proposed method to reconstruction-based adversarial attacks, we conduct an empirical study using a strong image-level reconstruction attacker based on SwinIR. Following standard practice in privacy-preserving face recognition, identity leakage is assessed using an external face recognizer, ArcFace. Under the ArcFace-based face verification protocol, the anonymized images yield an AUC of 0.633 and a TAR of 0.0383 at FAR = 0.01. After reconstruction, the performance becomes an AUC of 0.653 with a TAR of 0.0510 at FAR = 0.01. Although a marginal increase is observed, the absolute verification performance remains low, suggesting that reconstructed images do not allow reliable identity recovery. As shown in Fig. 6, reconstruction attacks do not produce visually more informative faces than the anonymized images, which is consistent with the quantitative results in Table X. Reconstruction attacks only



Fig. 6. Qualitative visualization of reconstruction attacks. For each subject, we show the original image, the anonymized image generated by our method, and the reconstructed image obtained using SwinIR (from left to right). The reconstructed images remain visually similar to the anonymized ones, indicating that reconstruction attacks do not recover identity-discriminative facial details.

TABLE X
FACE VERIFICATION PERFORMANCE UNDER RECONSTRUCTION ATTACKS

Evaluation Model	Input Type	AUC	TAR@FAR=0.01
ArcFace	Anonymized images	0.633	0.0383
ArcFace	Reconstructed images	0.653	0.0510
Identity Decoder (ours)	Reconstructed images	0.5154	0.0237

lead to marginal improvements in ArcFace-based verification performance, while the reconstructed images remain ineffective for identity decoding.

In addition, we analyze the impact of reconstruction attacks on the system itself by evaluating the reconstructed images using our identity decoder in inference mode. The decoder achieves an AUC of 0.5154 and a TAR of 0.0237 at FAR = 0.01, which is close to random performance, indicating that reconstruction attacks do not restore identity-discriminative information even for the decoding network.

F. Comparison With PPFR Methods

To provide a more comprehensive evaluation, we include representative PPFR methods in our experimental comparisons. This allows us to analysis the proposed method across both paradigms and to highlight the practical implications of preserving visually meaningful faces.

To rigorously evaluate the robustness of these models against extreme intra-class variations, we analyze the standard LFW [39] benchmark alongside two more challenging datasets: CPLFW [46] and CALFW [47], which focus on large pose variations and significant aging effects, respectively. We followed the standard evaluation protocols, utilizing 3,000 positive pairs and 3,000 negative pairs for each dataset. To ensure optimal performance for the identity recognizer, all images were pre-processed through facial alignment and resized to a standard 256×256 resolution. As shown in Table XI, while state-of-the-art PPFR methods such as DuetFace [48] and MinusFace [49] perform

TABLE XI
FACE VERIFICATION PERFORMANCE ON LFW, CPLFW AND CALFW DATASETS

Method	Venue	LFW	CPLFW	CALFW
InstaHide [50]	ICML'20	96.53	81.03	86.24
PEEP [51]	COSE'20	98.41	79.58	90.06
Cloak [52]	WWW'21	98.91	83.43	92.18
PPFR-FD [53]	AAAI'22	99.68	90.78	95.72
DuetFace [48]	ACM MM'22	99.82	92.35	96.10
DCTDP [54]	ECCV'22	99.77	91.37	96.05
PartialFace [55]	ICCV'23	99.80	92.03	96.07
MinusFace [49]	CVPR'24	99.78	91.90	95.59
Pro-Face C [15]	TIFS'24	99.82	93.75	95.87
Ours		100.00	100.00	100.00
Ours (JPEG Q=60)	-	99.71	99.63	99.62

exceptionally well on the standard LFW dataset, they often experience performance degradation when facing the drastic pose or age changes present in CPLFW and CALFW. In contrast, our Flex-Face framework achieves a perfect 100% verification accuracy consistently across all three benchmarks. To briefly illustrate the robustness of our active injection paradigm, we include the performance of Flex-Face under JPEG compression ($Q = 60$). Remarkably, its degraded performance still outperforms most baselines' clean performance. This exceptional consistency is attributed to our identity-aware embedding mechanism, which hierarchically injects encrypted identity cues into the global image manifold. By decoupling identity information from physical facial attributes, our method ensures that the identity recognizer remains invariant to non-identity-related geometric deformations and texture changes, even when the images are rendered into human-interpretable anonymous faces.

This exceptional consistency of high performance is attributed to our identity-aware embedding mechanism, which hierarchically injects encrypted identity cues into the global image manifold. Because our approach decouples identity from visual features by utilizing a highly robust 30-bit hidden identity message, the extraction of this identity message remains entirely unaffected by pose changes or aging effects. By decoupling identity information from physical facial attributes, our method ensures that the identity recognizer remains invariant to non-identity-related geometric deformations and texture changes, even when the images are rendered into human-interpretable anonymous faces. This explicit identity injection distinguishes our method from conventional PPFR, which suffers from poor visual quality of the generated face. Unlike many PPFR methods that rely on strong visual obfuscation to achieve privacy, IDFA maintains the structural and semantic integrity required for visual utility while ensuring robust machine recognition under complex real-world conditions.

Furthermore, we provide qualitative results in Fig. 7. As visualized, while the anonymized faces appear as entirely different individuals to protect visual privacy, our method precisely preserves the structural integrity and identity-aware cues. In contrast to most PPFR methods, our model reframes identity-preserving recognition as an identity embedding and decoding problem. This ensures that the identity recognizer remains invariant to



Fig. 7. Visualization of identifiable face anonymization results across different challenging datasets. Each column represents the same original identity. For each dataset pair, the top row is the original image, and the bottom row is the generated anonymized result. Results demonstrate high realism and identity change while successfully preserving non-biometric facial attributes (expression, pose, illumination) across standard (LFW), cross-age (CALFW), and cross-pose (CPLFW) conditions.

TABLE XII
HIERARCHICAL LAYER ABLATION ON CELEBA

Layers	LPIPS $\uparrow$	SSIM $\downarrow$	Rank1(%)	TAR(%)	BER
2	0.458	0.479	99.42	99.17	0.0103
4	0.474	0.483	100.00	100.00	0.0011
6	0.469	0.480	100.00	99.76	0.0023

non-identity-related geometric deformations (large pose variations in CPLFW) and texture changes (significant aging effects in CALFW), enabling strong visual anonymization with reliable recognition.

G. Ablation Study

To investigate the impact of hierarchical depth on identity retention, visual privacy, and robustness, we conducted an ablation study by varying the number of hierarchical injection layers from 2 to 4 and 6, while keeping all other settings unchanged. As shown in Table XII, shallow injection (2 layers) leads to insufficient propagation of identity-related information into deeper semantic feature levels, resulting in higher BER and slightly degraded recognition performance. Moderate hierarchical depth (4 layers) achieves the best overall trade-off, simultaneously maximizing identity recognition performance (Rank-1 and TAR), minimizing BER, and maintaining strong visual privacy (higher LPIPS). Excessive depth (6 layers) does not further improve identity retention and instead introduces feature redundancy and over-conditioning, which slightly degrades privacy metrics and increases BER.

From a representational perspective, hierarchical identity injection enables identity information to be progressively aligned with multi-scale facial semantics. However, injecting identity

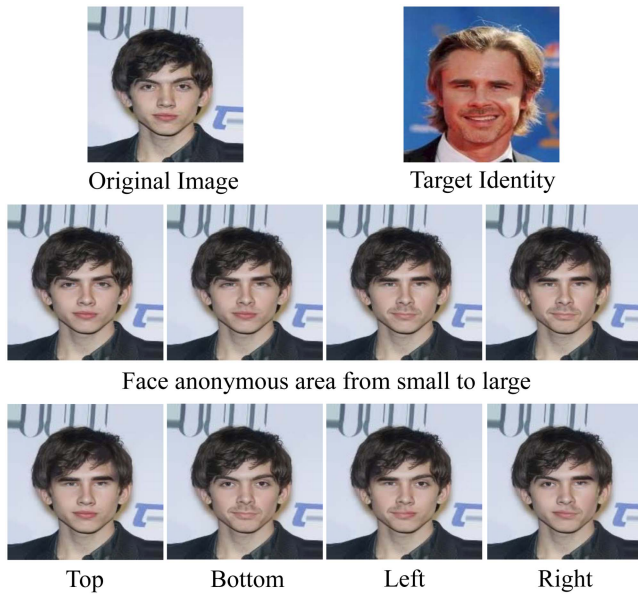


Fig. 8. Controllable facial privatization across different scales. The top row shows the original private image and our human-preferred target identity. The second row shows that the anonymous areas are changing from a small central area to large global areas. The bottom row illustrates the anonymous faces, which only modify the top/bottom/left/right areas.

embeddings at too many levels may interfere with anonymization objectives by over-constraining the feature space. Therefore, four hierarchical layers naturally emerge as a balanced design point, providing sufficient identity propagation while preserving visual privacy.

H. Facial Anonymization Across Different Scales

Our experiments demonstrate that it is feasible to anonymize varying proportions of the facial region, which explains why our Flex-Face is capable of effectively performing face swapping. Specifically, we incrementally increase the area subjected to face swapping until it covers the entire face. The results indicate that this process does not significantly affect the extraction of identity encodings, nor does it impact the face verification accuracy or Rank-1 recognition rate. As a result, we are able to achieve facial anonymization at any scale.

Fig. 8 presents the results of our arbitrary-scale facial anonymization. The first row displays the source and target images. The second row illustrates the gradual increase of the face-swapping area, ultimately resulting in the anonymized image. The third row demonstrates the swapping of half of the facial region; for example, in the case labeled "left," only the left half of the face is swapped, with the left half sourced from the target image and the right half from the original image.

These results demonstrate that our Flex-Face effectively supports customizable levels of facial anonymization, ranging from minor modifications to extensive transformations, while preserving selected facial regions. This flexibility enhances privacy protection while allowing users to maintain control over the extent and area of anonymization, showcasing the model's adaptability to varying privacy requirements.

TABLE XIII
ANONYMIZATION STABILITY UNDER VARIOUS TARGET DIFFERENCE LEVELS

Target Difference Level	SSIM(Source, Target)	SSIM(Source, Anonymized)
Low	0.16-0.30	0.895 ± 0.002
Medium	0.31-0.40	0.885 ± 0.005
High	0.41-0.47	0.877 ± 0.003

I. Diverse Generated Images

The innovation of our method lies in its ability to generate visually distinct and diverse anonymized images for the same identity as the original images. In contrast, other methods may create anonymized images that differ visually from the original images, but the anonymized images generated for the same identity are often quite similar. This is because our method diverges from those that directly extract facial features and compute similarities for identity recognition; instead, our approach relies on embedding identity encodings within the image itself. In our framework, the face swapping module is introduced primarily to generate perceptually realistic anonymized faces for human observation. It should be noted that this component is not essential to the proposed identity encoding mechanism. In principle, any anonymization operation (e.g., face swapping, synthetic generation, or even pixel-level perturbation) can be applied, as long as the identity encoding embedding process is preserved. Furthermore, our model employs a training strategy similar to MBRS, where the identity encoding is embedded through a spatial diffusion mechanism across multi-scale feature maps. Instead of concentrating the encoded bits in specific regions, the embedding process distributes identity information over the entire image. As a result, even when part of the image is removed, sufficient encoding redundancy remains for correct decoding. Additionally, while embedding identity encodings, it samples multi-scale, multi-channel embeddings, resulting in a near-zero bit error rate (BER) during identity encoding extraction.

In our experiments, the target identities are intentionally selected to be highly diverse from the source faces. Specifically, for each source image, we randomly select some target faces whose SSIM values with respect to the original image range from 0.19 to 0.47, indicating substantial visual and attribute-level differences. Despite this large variation in target identity appearance, the anonymized images consistently maintain high structural similarity to the original faces, with SSIM values in the range of 0.87–0.89. This observation suggests that the proposed method effectively decouples identity semantics from low-level facial attributes, allowing diverse human-preferred identities to be embedded without compromising visual fidelity. The quantitative analysis is summarized in Table XIII, which reports the structural similarity between the source faces and the anonymized results under various target difference levels. Representative visual examples are provided in Fig. 9. These results demonstrate that valid human-preferred identities in our framework are not restricted by visual similarity to the source face. Instead, the model constrains the influence of the target identity to identity-related information, while preserving the original facial structure and appearance, thereby supporting robust and flexible anonymization. The results of these diverse

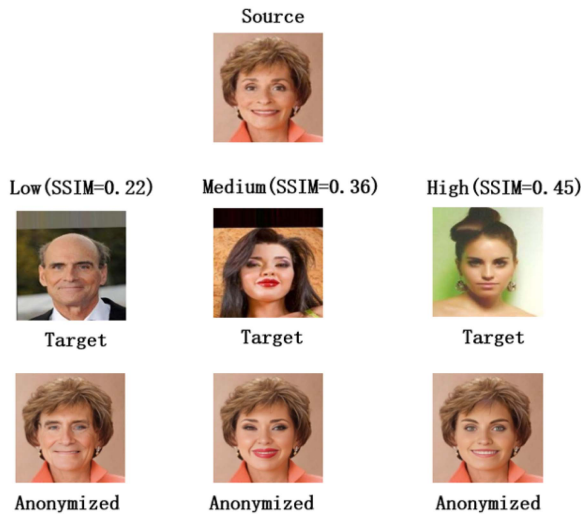


Fig. 9. Visualization of anonymization results under diverse human-preferred identities with varying target difference levels. The source face is fixed (top), while target identities with low, medium, and high SSIM differences are shown in the three columns. Despite substantial variation in target appearance, the anonymized images remain visually consistent with the source face.



Fig. 10. Visual results of the original private faces and corresponding diverse anonymous faces with various identities, which obtain good visual quality for human eyes as well as high recognition accuracy for machines.

anonymized images are shown in Fig. 10, where the first image in each row represents the original image, and the remaining images are the various anonymized versions generated from that original image. As shown in Fig. 11, the proposed method produces anonymized faces with more consistent facial structure and fewer visual artifacts compared to existing methods, while maintaining effective identity obfuscation. Moreover, for identity recognition and verification, we use the identity encoding as the identity information for the individual and compute the similarity between the embedded and extracted identity encodings. As a result, although the visual differences among anonymized images of the same person can be significant, the embedded identity encodings remain consistent, leading to a similarity close to 1. In contrast, the similarity between identity encodings of different individuals typically hovers around 0.6, which explains our high performance in recognition and verification tasks.



Fig. 11. Visual comparison with representative face anonymization.

V. CONCLUSION

This paper introduces a new paradigm to solve the privacy-utility paradox in face recognition, which ensures privacy protection with high visual quality, high recognition accuracy, flexibility, and controllability. Within this framework, a hierarchical identity injection method in conjunction with a privacy-consistent identity recognizer is proposed to incorporate identity information into facial images for achieving accurate identity recognition. An identity-consistent face anonymization method is employed to incorporate diverse human-preferred identities for face anonymization. Consequently, the proposed method provides a versatile, practical, and well-balanced solution for safeguarding visual privacy in a wide range of face recognition systems. Comprehensive experiments across diverse face image datasets and under various recognition and privacy-compromise scenarios have proven the superiority of this approach compared to other privacy-preserving face recognition and identifiable face anonymization techniques. To this end, Flex-Face aims to contribute to the development of more ethical and user-centric facial recognition technologies that respect individuals' privacy rights while maintaining high levels of recognition accuracy. We hope that our work may shed light on the ongoing exploration of enhancing controllability while simultaneously pursuing a good balance for privacy-protective face recognition.

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